

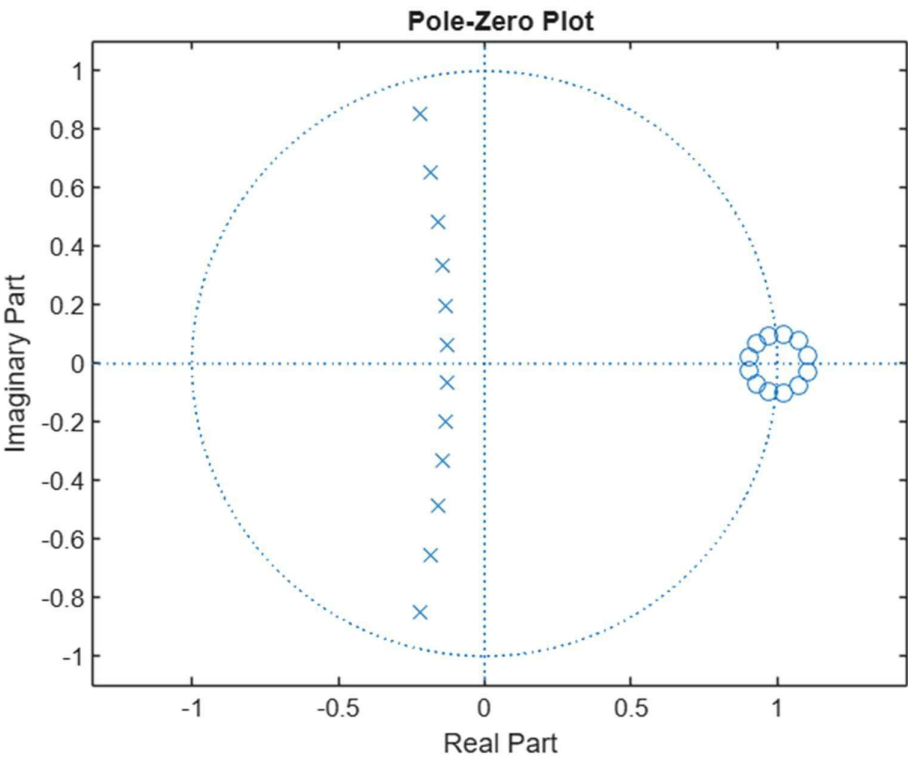
EX NO:1	Butterworth IIR Filters
DATE:	1.2. Design and Analysis of a Butterworth IIR High-Pass Filter Using MATLAB.

```

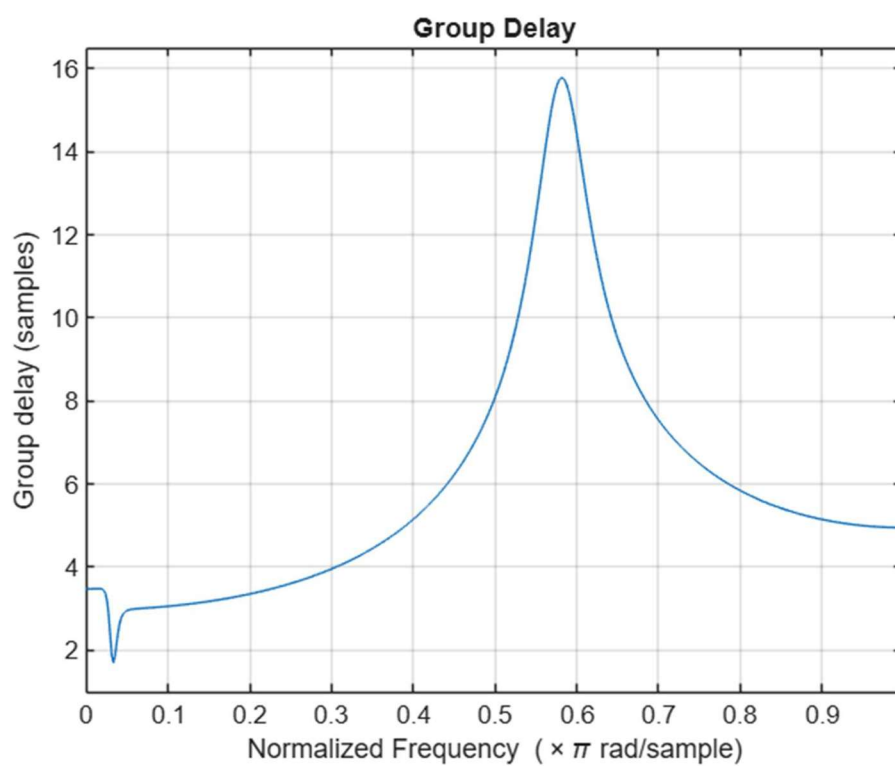
clc;
clear;
close all;
% Filter Specifications
Fs = 10000;    % Sampling Frequency (Hz)
Fp = 3000;     % Passband Frequency (Hz)
Fst = 2000;    % Stopband Frequency (Hz)
% Calculate Filter Order and Cutoff Frequency
[N, Wn] = buttord(Fp/(Fs/2), Fst/(Fs/2), 1, 60);
% Design High-Pass Butterworth Filter
[b, a] = butter(N, Wn, 'high');
% Frequency Response
figure;
freqz(b, a);
title('Frequency Response');
% Impulse Response
figure;
[h, n] = impz(b, a, 50);
stem(n, h, 'filled');
grid on;
title('Impulse Response');
xlabel('Samples');
ylabel('Amplitude');
% Step Response
figure;
stepz(b, a);
grid on;

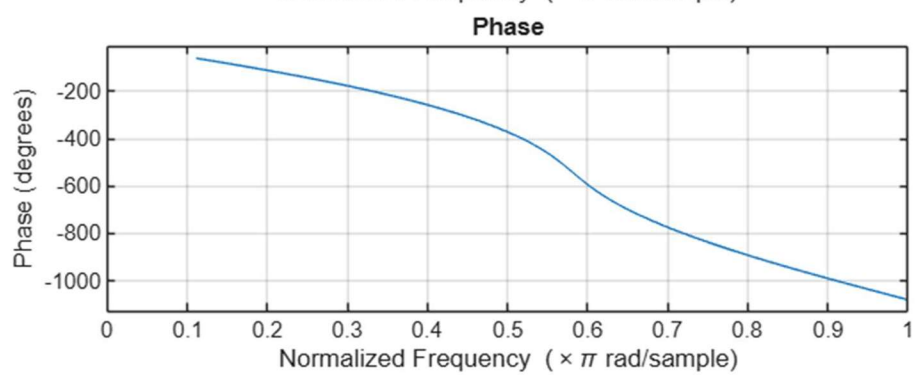
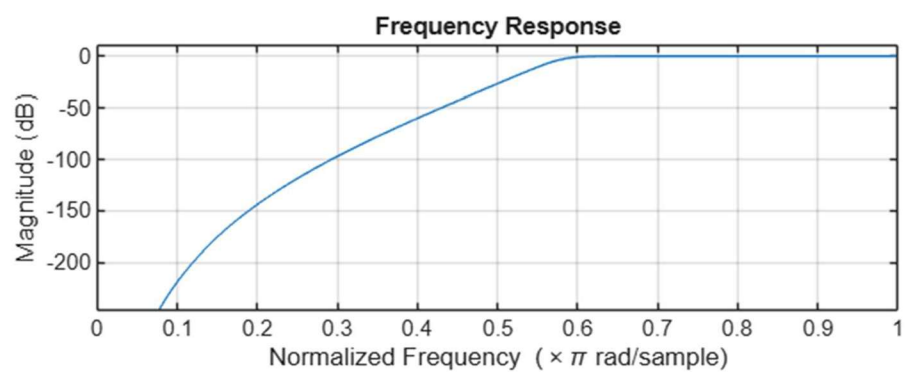
```

title('Step Response');

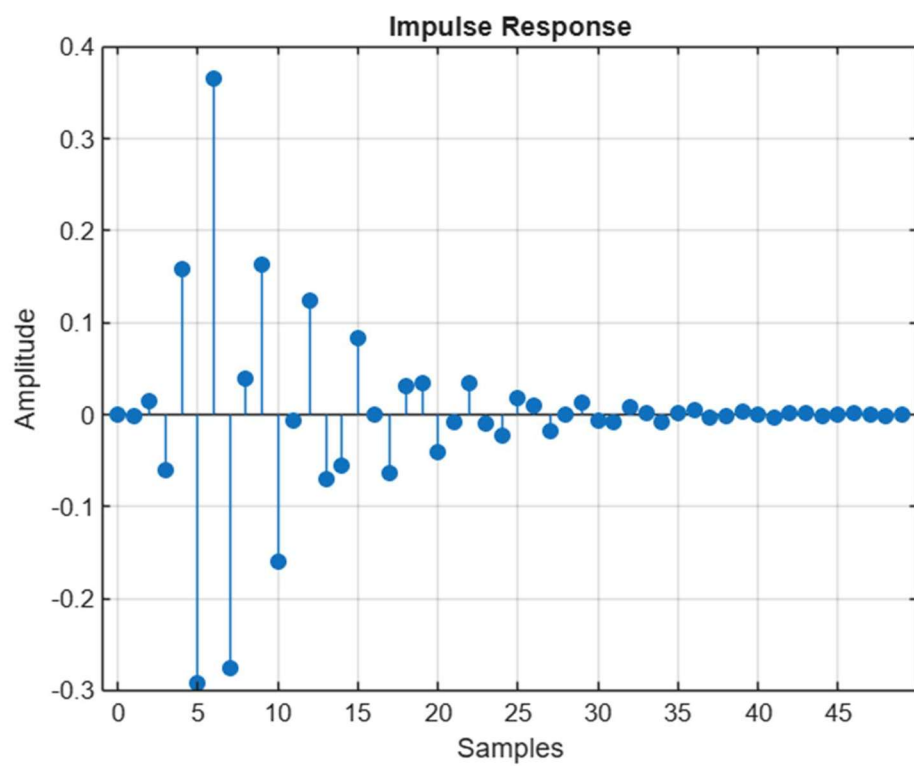


Minimum Filter Order (N) = 12

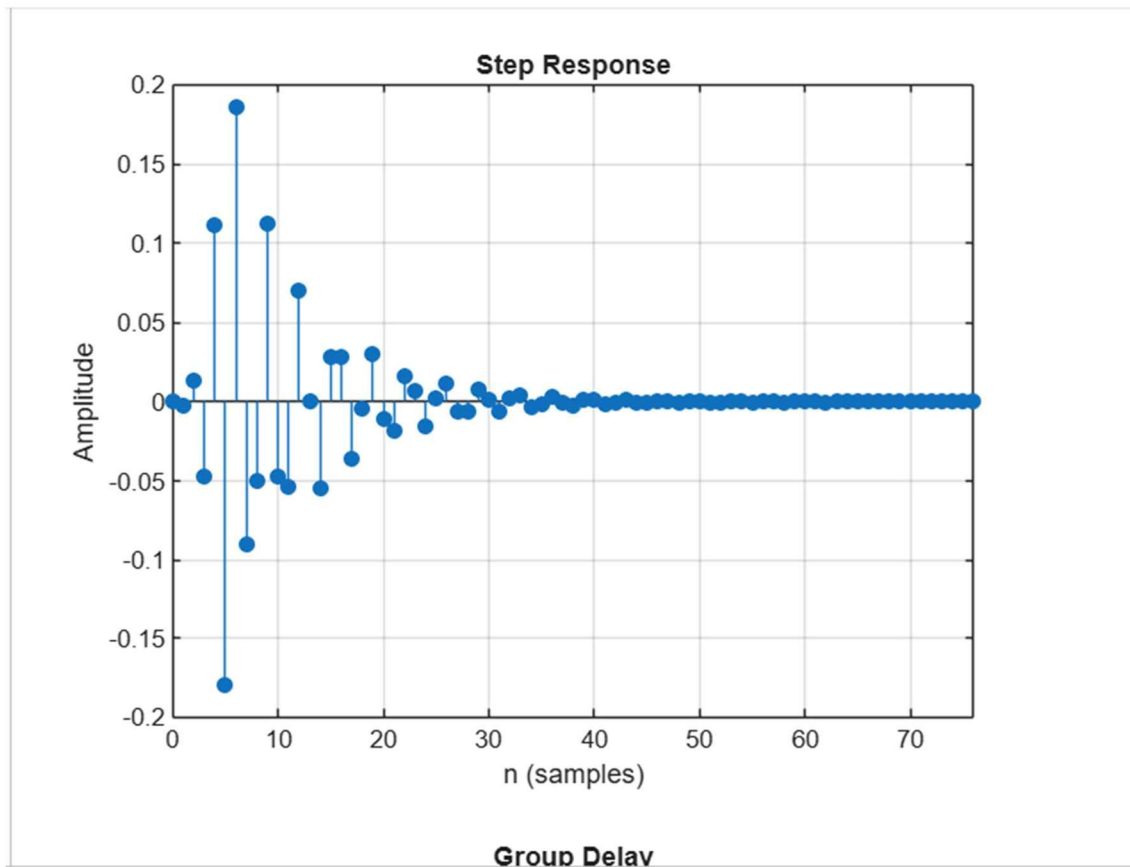




Impulse Response



Step Response



```
% Group Delay
```

```
figure;
```

```
grpdelay(b, a);
```

```
grid on;
```

```
title('Group Delay');
```

```
% Pole-Zero Plot
```

```
figure;
```

```
zplane(b, a);
```

```
title('Pole-Zero Plot');
```

```
% Display Filter Parameters
```

```
fprintf('Minimum Filter Order (N) = %d\n', N);
```

```
fprintf('Cutoff Frequency (Wn) = %.4f\n', Wn);
```

